EMUMADZ: Enhanced MUtation Mapping and Detection in Zebrafish

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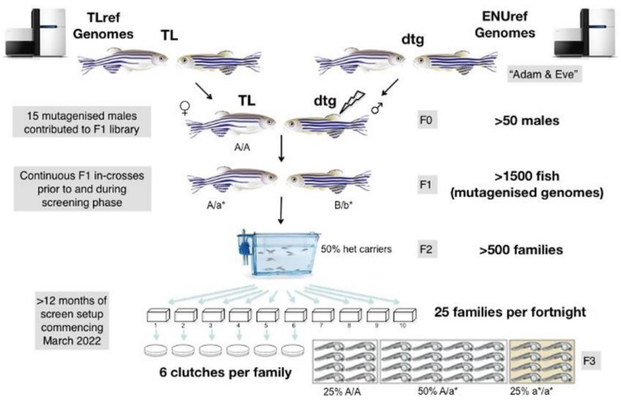
BACKGROUND

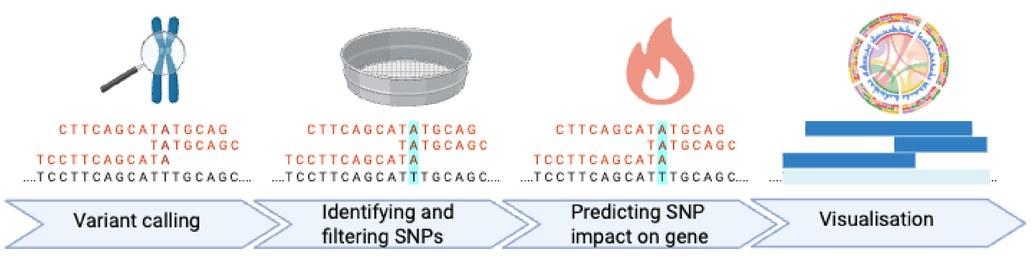
To discover new autosomal recessive mutations in biological processes of interest, we conducted a forward genetic screen on *Danio rerio*: the zebrafish.



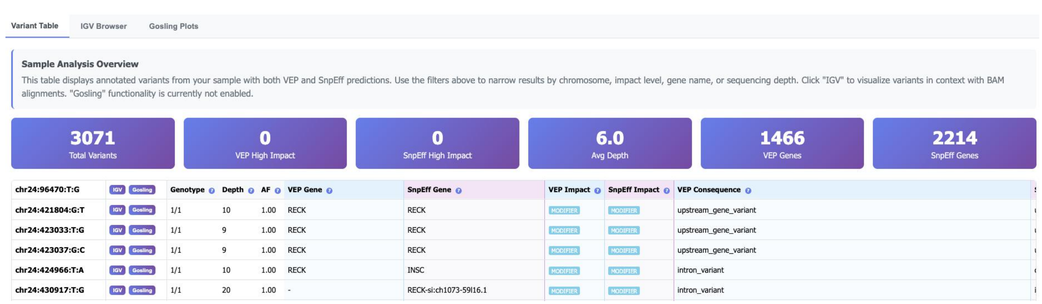
METHODS

1. Variant calling
2. SNP identification
3. Filtering strong candidate SNPs
4. Annotating SNP impact
5. Visualising interactive data

1. GATK
2. bcftools
3. ENSEMBL VEP and snpEff
4. IGV



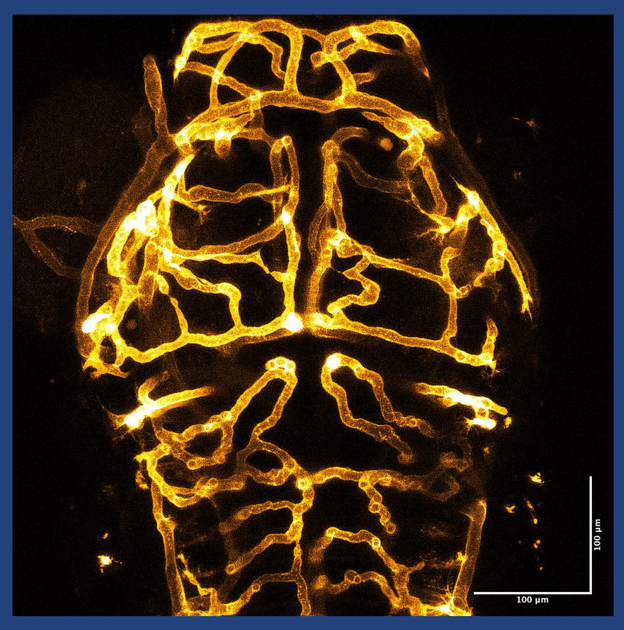
RESULTS



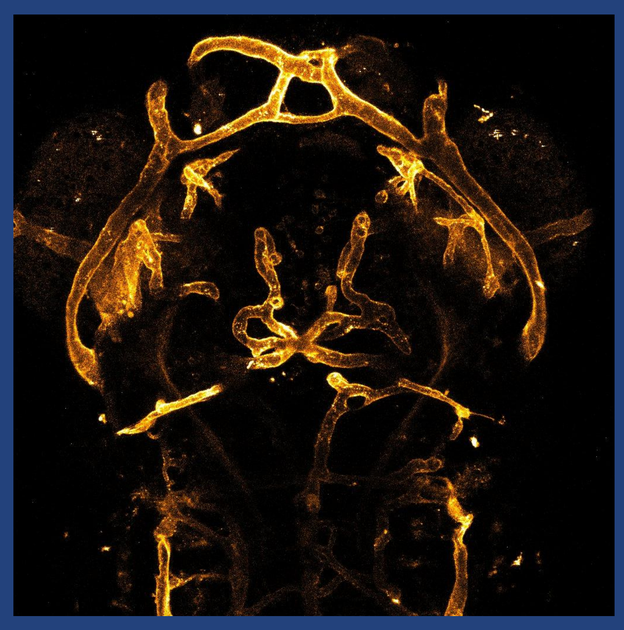
A subset of relevant information drawn from the variant calling and annotation pipeline are provided in an **interactive table**. We display homozygosity/heterozygosity states of the mutant samples and the predicted SNP impact to help narrow down candidate mutants.

Our **interactive genome browser** allows users to explore samples. We detect a homozygous SNP in the mutant sample which is absent in the Wild-Type control. The impact of the SNP was computationally predicted to be high & this hypothesis is supported by the displayed phenotype.

We discovered a novel **SNP in RECK** with an expected phenotype: **a lack of major blood vessels in the brain.**



Wild-Type sample. Blood vessel density in the brain is normal. Scale bar is identical for both images. *Image provided by Oguzhan Baltaci.*



Mutant RECK sample. Blood vessel density in the brain is limited. Scale bar is identical for both images. *Image provided by Oguzhan Baltaci.*

EXTRA FIGURES / DATA

Detailed, readable documentation:
- for both biologists
- and `bioinformaticians`

Code follows `FAIR` principles:
- version controlled repository
- `docker/apptainer` images for portability
- open source software license

Additional features in progress:
- genome-wide plots showing mutation hotspots
- `nextflow` pipeline
- `gosling` advanced visualisation

CONCLUSION

We optimised a pipeline that (a) mapped mutations to a region of genomic linkage and (b) identified candidate SNPs predicted to be damaging to gene function, matching (c) essential criteria of homozygous mutations which are ENU-induced and absent in control reference samples.

The high-level API is designed to provide easy usage for individuals running the pipeline. Code is designed in a modular way, allowing users to reuse or modify specific processes to suit their own needs. Detailed documentation exists, with two case studies on *Danio rerio*: the zebrafish. One case study covers the comparison of the F0-F2 [wild-type grandparent:mutant grandchildren] generation, and the second case study covers the comparison of F2-F2 [wildtype sibling:mutant sibling] generation.

Individual software tools have been designed for forward genetic screens, but most tools and higher-level pipelines have (a) an organism bias towards humans and mice and (b) are designed for addressing specific aspects of SNPs instead of the SNP characterisation process as a whole.

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